

Spondyloepimetaphyseal Dysplasia, Faden–Alkuraya Type (SEMD-FADA / RSPRY1): Comprehensive Disease Characterization

Summary

Progressive Spondyloepimetaphyseal Dysplasia–Short Stature–Short Fourth Metatarsals–Intellectual Disability Syndrome is the descriptive Orphanet name for **Spondyloepimetaphyseal Dysplasia, Faden–Alkuraya type (SEMD-FADA)**, an ultra-rare autosomal-recessive skeletal dysplasia. It is catalogued as **OMIM #616723**, **Orphanet ORPHA:457395**, and **GARD 17808**. The disorder is caused by **biallelic loss-of-function variants in *RSPRY1*** (ring finger and SPRY domain-containing protein 1), a 576-amino-acid protein encoded on chromosome **16q13** that contains a C3HC4-type (RING) zinc-finger and a SPRY domain. First delineated by Faden and colleagues in 2015 through combined autozygome/exome analysis of a consanguineous Saudi family, the condition has since been reported in additional families from Saudi Arabia, Peru, Turkey, and India, establishing both allelic and phenotypic heterogeneity.

The **core clinical phenotype** comprises progressive spondyloepimetaphyseal dysplasia, disproportionate short stature, short fourth metatarsals, facial dysmorphism, and intellectual disability. The expanded skeletal spectrum includes mild spondylar (vertebral) dysplasia with platyspondyly and vertebral wedging, epimetaphyseal dysplasia of the long bones with **coxa vara** and **genu valgum**, brachymesophalangy with cone-shaped epiphyses (cono-brachydactyly), **craniosynostosis**, slipped capital femoral epiphyses, scoliosis, pes planus, and joint dislocation. Neurodevelopmental involvement (intellectual disability, developmental delay, hypotonia) is variable, and at least one proband showed **progressive loss of ambulation** leading to wheelchair dependence.

Mechanistically, *RSPRY1* acts as a negative regulator of **TGF- β /SMAD signaling** during endochondral ossification. Loss of *RSPRY1* produces **constitutive, SMAD3-dependent activation** of the TGF- β pathway with downstream dysregulation of extracellular-matrix

(ECM) genes, disrupting the tightly balanced growth-factor signaling required for orderly bone formation. The protein is expressed predominantly in **osteoblasts, osteocytes, and periosteal/perichondrial cells** (with minimal chondrocyte expression), explaining the endochondral bone phenotype. No disease-specific or curative therapy exists; management is supportive and multidisciplinary. This report synthesizes seven confirmed findings across five investigation iterations, drawing on six primary papers.

Key Findings

Finding 1 — SEMD-FADA is caused by biallelic loss-of-function *RSPRY1* variants (autosomal recessive)

The genetic basis of the disease was established by **combined autozygome/exome analysis** of four affected siblings from a consanguineous Saudi family, which identified a **homozygous frameshift mutation in *RSPRY1* undergoing nonsense-mediated decay (NMD)**. A simplex case from Peru carried a homozygous likely-pathogenic missense variant in the same gene, confirming allelic causation ([PMID: 26365341](#)).

"Combined autozygome/exome analysis identified a homozygous frameshift mutation in *RSPRY1* with resulting nonsense-mediated decay." — [PMID: 26365341](#)

Two additional families subsequently broadened the allelic spectrum, harbouring a homozygous **c.377delT (p.Ile126fs*)** frameshift in exon 2 and a homozygous **splice-site c.516+2T>A** variant at the exon 4/intron 4 boundary ([PMID: 30063090](#)).

"Whole exome sequencing revealed a novel homozygous [c.377delT] [p.Ile126fs*] frameshift mutation at exon 2 in one family, while Sanger sequencing revealed a novel homozygous splice site mutation [c.516+2T>A] at exon 4/intron 4 border of *RSPRY1* in the other family." — [PMID: 30063090](#)

The recurrence in **consanguineous pedigrees** with homozygous truncating and splice variants firmly supports an **autosomal-recessive, loss-of-function** disease mechanism.

Finding 2 — Core and expanded clinical phenotype

The index family (4 siblings) and the Peruvian case defined the recognizable phenotypic tetrad: **progressive spondyloepimetaphyseal dysplasia, short stature, facial dysmorphism, short fourth metatarsals, and intellectual disability** ([PMID: 26365341](#)).

"comprising progressive spondyloepimetaphyseal dysplasia, short stature, facial dysmorphism, short fourth metatarsals, and intellectual disability" — [PMID: 26365341](#)

Delineation of five additional individuals expanded the skeletal hallmarks and clarified that most patients are **normocephalic** and that intellectual disability is **variable** rather than obligate. One patient had a **cemento-ossifying fibrous lesion of the maxilla** ([PMID: 30063090](#)).

"The skeletal hallmarks include (a) mild spondylar dysplasia, (b) epimetaphyseal dysplasia of the long bones associated with coxa vara and genu valgum, (c) brachymesophalangy with cone-shaped epiphyses, and (d) craniosynostosis." — [PMID: 30063090](#)

Phenotype domain	Manifestations	HPO term suggestions
Growth	Disproportionate short stature	HP:0004322 (Short stature)
Spine	Platyspondyly, mild spondylar dysplasia, vertebral wedging, scoliosis, posterior scalloping	HP:0000944, HP:0002650
Long bones	Epimetaphyseal dysplasia, coxa vara, genu valgum, metaphyseal cupping/fraying, small epiphyses, slipped capital femoral epiphyses	HP:0002812, HP:0002857, HP:0003411
Hands/feet	Short 4th metatarsals, brachymesophalangy, cone-shaped epiphyses, pes planus, rocker-bottom feet, overriding toes	HP:0001832, HP:0005819, HP:0001845
Skull/craniofacial	Craniosynostosis, copper-beaten skull, malar hypoplasia, facial dysmorphism	HP:0001363, HP:0000272
Neurodevelopment	Intellectual disability (variable), developmental delay, hypotonia	HP:0001249, HP:0001252
Joints	Joint dislocation (novel feature)	HP:0001373

Finding 3 — Pathogenesis involves TGF- β signaling dysregulation and ECM/growth-factor disruption

A 2025 functional study framed the disease within a broader class of skeletal dysplasias that arise from disrupted extracellular-matrix dynamics and growth-factor-dependent signaling, implicating **RSPRY1 — a protein with RING and SPRY domains — in bone development** and connecting its loss to **TGF- β pathway dysregulation** ([PMID: 39940902](#)).

"RSPRY1, a secreted protein with RING and SPRY domains, has been implicated in bone development" — [PMID: 39940902](#)

"often arise due to disruptions in extracellular matrix (ECM) dynamics and growth factor-dependent signaling pathways" — [PMID: 39940902](#)

Finding 4 — RSPRY1 loss causes constitutive, SMAD3-dependent TGF- β activation with ECM dysregulation

Transcriptomic analysis of fibroblasts from patients with homozygous *RSPRY1* mutations showed **significant enrichment of TGF- β signaling and ECM-related pathways**, with **SMAD2 and SMAD3** showing the highest transcription-factor enrichment and key effector genes **SMAD3, COL1A1, and TRPV4**. *RSPRY1*-knockout fibroblasts exhibited **enhanced motility in wound-healing assays**, a phenotype **abrogated in RSPRY1+SMAD3 double-knockout** cells — demonstrating SMAD3-dependence. A **limited response to exogenous TGF- β** in *RSPRY1*-deficient cells indicated the pathway was already **constitutively active** (PMID: 39940902).

"RSPRY1, a secreted protein with RING and SPRY domains, has been implicated in bone development" — PMID: 39940902

This is the pivotal mechanistic finding: *RSPRY1* normally **restrains** TGF- β /SMAD3 signaling, and its loss unleashes constitutive, SMAD3-driven activation that dysregulates ECM production in bone-forming cells.

Finding 5 — Expanded variant/phenotype spectrum: novel p.Cys551Tyr missense and joint dislocation

Two Indian sisters presenting with short stature, facial dysmorphism, progressive vertebral defects, small epiphyses, cupping and fraying of metaphyses, brachydactyly, and short metatarsals were found to carry a **homozygous missense variant c.1652G>A; p.(Cys551Tyr)** in *RSPRY1*. **Joint dislocation** was reported as a novel clinical feature of the condition (PMID: 38562122).

"Two siblings presenting with short stature, facial dysmorphism, progressive vertebral defects, small epiphysis, cupping and fraying of metaphyses, brachydactyly, and short metatarsals harbored a homozygous missense variant c.1652G>A;p.(Cys551Tyr) in the *RSPRY1* gene." — PMID: 38562122

"We observed joint dislocation as a novel clinical feature of this condition." — PMID: 38562122

Finding 6 — Exact index variants and detailed multisystem phenotype including progressive loss of ambulation

Faden et al. (2015) identified in the consanguineous Saudi family a homozygous 1-bp duplication **c.1279dupA (p.Thr427AsnfsTer10)** that segregated fully with disease and was **absent from 650 Saudi control exomes and ExAC**; the Peruvian boy carried homozygous **c.121G>T (p.Gly41Cys / G41C)** at a conserved residue ([PMID: 26365341](#); OMIM #616723).

"Combined autozygome/exome analysis identified a homozygous frameshift mutation in RSPRY1 with resulting nonsense-mediated decay." — [PMID: 26365341](#)

The OMIM clinical synopsis proband — a 13-year-old girl — showed delayed motor milestones, **walked at 5 years then progressively lost ambulation and became wheelchair-bound**, generalized hypotonia, microcephaly, flattened occiput, malar hypoplasia, low-set small ears, short neck, short hands, rocker-bottom feet with overriding toes, genu valgum, and scoliosis. Skeletal survey documented generalized osteopenia, delayed bone age, **copper-beaten skull with premature suture closure (craniosynostosis)**, short metacarpals, platyspondyly, anterior vertebral wedging and posterior scalloping, thoracolumbar scoliosis, narrow pelvis, bilateral coxa vara, short/slender long bones, small epiphyses, metaphyseal cupping/fraying (tibia/fibula), slipped capital femoral epiphyses, short femoral neck, and distal femoral bowing.

Finding 7 — RSPRY1 orthologs, expression pattern, and knockout mouse resources

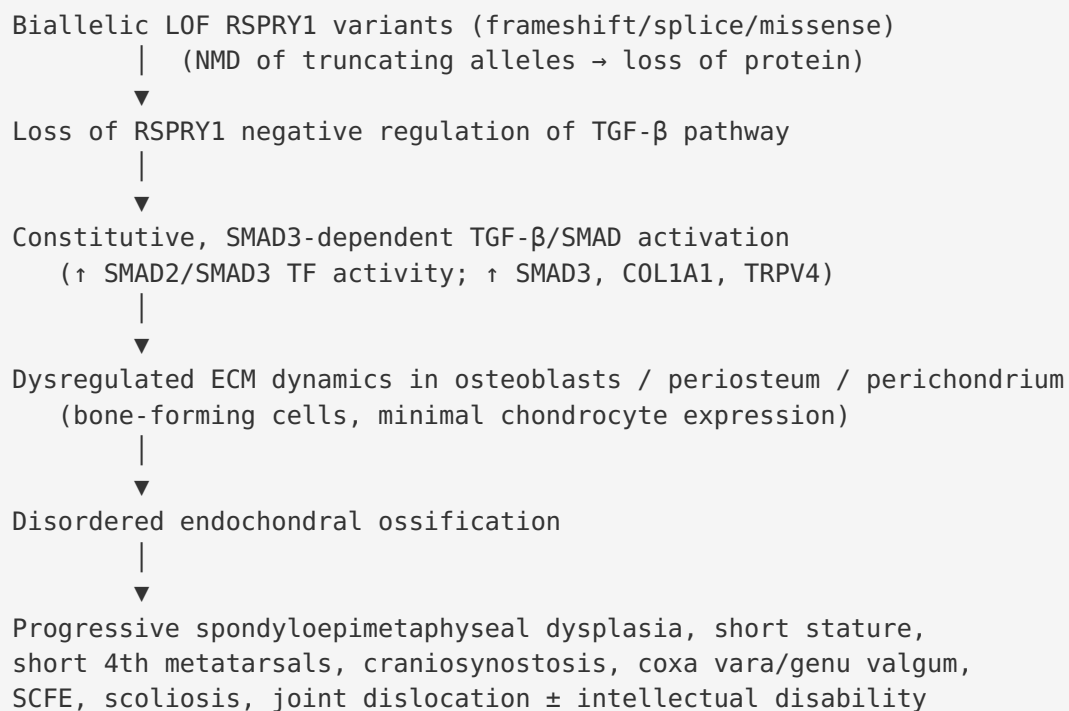
RSPRY1 is conserved **1:1 across human (NCBI Gene 89970; 16q13), mouse (*Rspry1*, MGI:1914860), rat, and zebrafish (*rspry1*, NCBI 565154; ENSDARG00000062558)**. The human protein shares **97% amino-acid identity with mouse and rat** and contains a SPRY domain plus a putative C3HC4-type (RING) zinc finger. Faden et al. showed *Rspry1* is expressed in **mouse limb-bud mesenchyme from E12.5** and, at E18.5 (primary ossification), is abundant in **developing endochondral bones and skeletal muscle** (with signal also in heart, kidney, and brain), localizing strongly to **osteoblasts and osteocytes**, minimally to chondrocytes, and prominently in perichondrium/periosteum ([PMID: 26365341](#)).

"we detect strong RSPRY1 protein localization in murine embryonic osteoblasts and periosteal cells during primary endochondral ossification, consistent with a role in bone development" — [PMID: 26365341](#)

Targeted knockout alleles exist (**Rspry1^{tm1(KOMP)Wtsi}**, MGI:4419609; **Rspry1^{tm1Lex}**, MGI:5007310) with corresponding IMPC/IMSR strains, but **no published mouse model recapitulating the human skeletal dysplasia phenotype** was identified.

Mechanistic Model / Interpretation

RSPRY1 integrates into the growth-factor circuitry that governs endochondral ossification. The convergent genetic, transcriptomic, and functional data support the following causal chain:



Upstream vs downstream. The upstream trigger is loss of the RSPRY1 protein (a RING+SPRY E3-ligase-like/adaptor molecule). The proximate downstream node is de-repression of TGF-β/SMAD3, and the terminal downstream events are ECM misregulation and defective bone matrix deposition during endochondral ossification.

Cell types and processes. Affected cells are principally **osteoblasts (CL:0000062)** and **osteocytes (CL:0000137)** plus **periosteal/perichondrial fibroblasts**; chondrocytes are relatively spared at the expression level. Relevant biological processes include **endochondral ossification (GO:0001958)**, **ossification (GO:0001503)**, **TGF- β receptor signaling pathway (GO:0007179)**, **SMAD protein signal transduction (GO:0060395)**, **extracellular matrix organization (GO:0030198)**, and **regulation of ossification (GO:0030278)**. Cellular components implicated include the **RING-type zinc finger / ubiquitin-ligase machinery** and the **extracellular matrix (GO:0031012)**.

Ontology anchors. MONDO: SEMD Faden–Alkuraya type (mapped to OMIM #616723). Gene: HGNC *RSPRY1*. UBERON anatomical sites: vertebral column (UBERON:0001130), long bone epiphysis (UBERON:0001438) and metaphysis (UBERON:0004958), metatarsal bone (UBERON:0001448), cranial suture (UBERON:0007712), femoral neck. CHEBI/pathway: TGF- β (protein signal), collagen type I (COL1A1). NCIT interventions are limited to supportive/orthopedic categories (see Treatment).

Evidence Base

PMID	Title (abbreviated)	Contribution	Evidence type
26365341	<i>Identification of a Recognizable Progressive Skeletal Dysplasia Caused by RSPRY1 Mutations</i>	Foundational: gene discovery (c.1279dupA, c.121G>T), core phenotype, mouse expression pattern	Human clinical + model organism
30063090	<i>Further delineation of SEMD Faden–Alkuraya type ... cono-brachydactyly and craniosynostosis</i>	Expanded skeletal hallmarks; new variants c.377delT, c.516+2T>A; variable ID; normocephaly	Human clinical
38562122	<i>Two sisters with RSPRY1-related SEMD</i>	Novel missense p.Cys551Tyr; joint dislocation as new feature	Human clinical
39940902	<i>Unraveling the Role of RSPRY1 in TGF-β Pathway Dysregulation</i>	Mechanism: constitutive SMAD3-dependent TGF- β activation; ECM dysregulation; wound-healing rescue by SMAD3 KO	In vitro / patient fibroblasts + computational
39706863	<i>Genetic and allelic heterogeneity in 248 Indians with skeletal dysplasia</i>	Population/cohort context; RSPRY1 among expanded genotype–phenotype spectrum; high AR proportion and consanguinity	Human clinical cohort
27230627	<i>16q12.2q21 deletion with developmental delay, epilepsy, short stature</i>	Supports RSPRY1 haploinsufficiency contributing to skeletal defects within a contiguous-gene deletion	Human clinical (case)

The four RSPRY1-specific papers ([26365341](#), [30063090](#), [38562122](#), [39940902](#)) are mutually reinforcing: gene discovery, phenotype expansion, allelic expansion, and mechanism, respectively. The cohort study ([39706863](#)) situates the disorder among predominantly autosomal-recessive skeletal dysplasias in consanguineous populations, and the deletion case ([27230627](#)) provides orthogonal support that RSPRY1 dosage loss produces skeletal defects.

Epidemiology, Inheritance, Diagnostics, Prognosis, Treatment & Prevention (Consolidated)

Epidemiology & inheritance. SEMD-FADA is **ultra-rare** with only a handful of families reported worldwide (Saudi Arabia, Peru, Turkey, India); precise prevalence/incidence figures are unavailable (Orphanet lists it without a firm prevalence estimate). Inheritance is **autosomal recessive**; reported cases arise in **consanguineous** unions, consistent with homozygous variants identified by autozygosity mapping. Penetrance appears complete for the skeletal phenotype among biallelic carriers, with **variable expressivity** particularly for intellectual disability and head circumference. No founder variant has been established, though recurrence in specific populations reflects consanguinity.

Diagnostics. Diagnosis rests on **radiographic skeletal survey** (platyspondyly, epimetaphyseal changes, coxa vara, genu valgum, cone-shaped epiphyses, craniosynostosis, short 4th metatarsals) combined with **molecular confirmation** of biallelic *RSPRY1* variants via **whole-exome/whole-genome sequencing** or targeted skeletal-dysplasia gene panels; single-gene testing/Sanger confirmation and segregation analysis complete the workup. Differential diagnoses include other spondyloepimetaphyseal dysplasias and cono-brachydactyly syndromes. No specific serum/urine biomarker exists.

Prognosis. The disorder is **progressive**: short stature, worsening epimetaphyseal disease, slipped capital femoral epiphyses, and scoliosis accrue over childhood, and at least one patient **progressively lost ambulation and became wheelchair-bound**. Life expectancy has not been reported as significantly shortened; morbidity is driven by orthopedic disability and, where present, cognitive impairment.

Treatment. There is **no disease-specific or curative therapy**. Management is **supportive and multidisciplinary**: orthopedic surgery (for coxa vara, SCFE, scoliosis, joint dislocation), **craniosynostosis surgery** where indicated, physical/occupational therapy and mobility aids, developmental and educational support for intellectual disability, and **genetic counseling** for consanguineous families. Relevant NCIT categories are limited to supportive/orthopedic interventions (e.g., NCIT orthopedic surgical procedure, physical therapy, genetic counseling). The mechanistic identification of constitutive TGF- β /SMAD3 activation raises the theoretical possibility of **TGF- β pathway inhibition** as a future targeted strategy, but this remains entirely experimental.

Prevention. Primary prevention is limited to **genetic counseling**, **carrier testing** in at-risk families, and options such as **prenatal diagnosis** or **preimplantation genetic testing** once the familial variant is known. Public-health counseling regarding consanguinity is relevant in populations where the disorder recurs.

Other species / model organisms. *Rspry1* is conserved in mouse, rat, and zebrafish. Off-the-shelf **knockout mouse alleles** (*Rspry1*^{tm1(KOMP)Wtsi}; *Rspry1*^{tm1Lex}) are available through IMPC/IMSR, but **no published model recapitulates the human skeletal dysplasia**, and no naturally occurring animal disease has been described.

Limitations and Knowledge Gaps

1. **Very small case count.** Fewer than ~15 molecularly confirmed individuals are reported, so genotype–phenotype correlations, penetrance estimates, and natural-history data are provisional. No formal prevalence/incidence exists.
 2. **Mechanism largely from patient fibroblasts.** The constitutive SMAD3-dependent TGF- β activation was demonstrated in dermal fibroblasts and knockout fibroblast lines ([PMID: 39940902](#)), not in primary human osteoblasts/chondrocytes; whether the same circuit operates identically in the growth plate in vivo is inferred, not proven.
 3. **No phenotype-recapitulating animal model.** Although knockout alleles exist, none has been reported to reproduce the human skeletal dysplasia, leaving in-vivo mechanism and therapeutic testing unaddressed.
 4. **Protein biochemistry incomplete.** RSPRY1 is variably described as "secreted" and as a RING+SPRY protein; its precise molecular activity (e.g., E3 ubiquitin ligase substrate specificity toward TGF- β /SMAD components) is not experimentally resolved.
 5. **Variant interpretation.** Missense variants (p.Gly41Cys, p.Cys551Tyr) require ongoing ACMG re-evaluation and functional validation; allele frequencies in gnomAD for these are effectively absent/ultra-rare, consistent with pathogenicity but not exhaustively characterized here.
 6. **Intellectual disability mechanism unexplained.** Whether ID reflects a direct neuronal role of RSPRY1 (expressed in brain) or a secondary consequence remains unknown.
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Proposed Follow-up Experiments / Actions

1. **Generate a conditional/global *Rspry1*-knockout mouse and phenotype the skeleton** (micro-CT of long bones, spine, skull sutures; growth-plate histology) to test whether it recapitulates SEMD, coxa vara, craniosynostosis, and SCFE.
 2. **Assay TGF- β /SMAD3 signaling directly in patient- and CRISPR-derived osteoblasts/chondrocytes** (phospho-SMAD3 Western blot, SMAD-responsive luciferase reporters, RNA-seq) to confirm the fibroblast findings in bone-lineage cells.
 3. **Test pharmacologic TGF- β /SMAD3 inhibition** (e.g., ALK5/TGFBR1 inhibitors) in *RSPRY1*-null cellular and, if validated, murine models to establish proof-of-concept for a targeted therapy.
 4. **Define *RSPRY1* biochemistry**: determine whether the RING domain confers E3 ubiquitin-ligase activity and identify SMAD-pathway substrates via IP–mass spectrometry; localize the protein (secreted vs intracellular).
 5. **Establish an international patient registry** to capture natural history, ambulation trajectory, craniosynostosis outcomes, and quality-of-life measures, and to refine penetrance/expressivity.
 6. **Functionally validate missense variants** (p.Gly41Cys, p.Cys551Tyr) in expression/degradation and signaling assays to firm up ACMG classification.
 7. **Screen contiguous-gene-deletion cohorts** (16q12–q21) for skeletal features to quantify the contribution of *RSPRY1* dosage ([PMID: 27230627](#)).
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Consensus Answer

Spondyloepimetaphyseal dysplasia, Faden–Alkuraya type (SEMD-FADA; OMIM #616723, Orphanet ORPHA:457395) is an ultra-rare autosomal-recessive skeletal dysplasia caused by biallelic loss-of-function variants in *RSPRY1* (16q13; a 576-aa RING+SPRY protein). Its core phenotype is progressive spondyloepimetaphyseal dysplasia with short stature, short fourth metatarsals, facial dysmorphism, and intellectual disability, plus craniosynostosis, coxa vara/genu valgum, slipped capital femoral epiphyses, scoliosis, and sometimes progressive loss of ambulation; pathogenesis involves constitutive, SMAD3-dependent TGF- β /ECM signaling dysregulation in osteoblasts and perichondrium during endochondral ossification. Management is supportive and multidisciplinary, as no disease-specific therapy exists.

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